The National Hospital for Neurology and Neurosurgery

POST-OPERATIVE / POST-PROCEDURAL OBSERVATIONS

Frequency of observations in the post-operative period should be maintained regardless of the clinical area to which the patient is transferred. If below standards cannot be met then this must be escalated to senior staff

University College London Hospitals NHS Foundation Trust

FREQUENCY of POST-OPERATIVE OBSERVATIONS

Every 20 minutes for **2 hours**
½ hourly for **2 hours**
1 hourly for at least **12 hours**
(or until 0800 following morning)
Reduce to 2 hourly if condition allows

If patient condition causing concern at any time increase frequency of observations and inform nurse in charge / medical staff

Following **INTRACRANIAL PROCEDURES**:
- GCS
- PUPILARY RESPONSE
- LIMB ASSESSMENT
- VITAL SIGNS

Following **SPINAL PROCEDURES**:
LIMB ASSESSMENTS:
- ***C1-T6*** *– upper & lower limbs*
- ***T7*** *&* ***below*** *– lower limbs only*
- Mullane Chart observations

VITAL SIGNS

POST-OPERATIVE ALERTS in EPIC

SHOUT ALERT following **A**nterior **C**ervical **D**iscectomy (ACD)

S = **S**welling of neck
H = **H**oarseness / voice changes
O = **O**esophageal discomfort / swallowing difficulty / drooling
U = **U**nusual behaviour / agitation
T = **T**achypnoea / difficulty breathing

! This patient has had anterior neck surgery !	
S	Swelling of neck
H	Hoarseness / voice changes
O	Oesophageal dysphagia / drooling
U	Unusual behaviour / agitation
T	Tachypnoea / difficulty breathing

If any of these symptoms are present contact **Neurosurgical SpR immediately**

WARNING ALERT following **C**arotid **E**ndarterectomy (CEA)

W = **W**ound swelling /bleeding
A = **A**irway compromise / breathing difficulties / tachypnoea / hoarseness / dysphagia
R = **R**hythm changes / heart rate changes
N = **N**eurological deterioration / confusion / seizure
I = **I**ncreasing / decreasing BP outside specified post-operative parameters
N = **N**ew or increasing headache unresolved with analgesia
G = **G**et immediate review by medical team if any of above signs or symptoms present

CAROTID ENDARTERECTOMY
Post-operative signs and symptoms which are a ***WARNING*** *of possible complications and require immediate* ***REVIEW***

W	Wound / neck swelling / bleeding
A	Airway compromise / breathing difficulties / tachypnoea / hoarseness / voice changes / dysphagia
R	Rhythm changes / heart rate changes
N	Neurological deterioration / confusion / seizure
I	Increasing / decreasing **BP** outside specified post-operative parameters
N	New or increasing headache unresolved with analgesia
G	Get immediate **REVIEW** by medical team if any of above signs or symptoms present

MEDICAL DETERIORATION	Stroke team (bleep 1146/8158) + UCH Vascular team (bleep 5026) + Outreach (bleep 8277) + Anaesthetic SpR (1st on call bleep 8131)
WOUND COMPLICATIONS	NHNN NV Surgical team (bleep 2137 / 8244) + UCH Vascular team (bleep 5026) + Outreach (bleep 8277) + Anaesthetic SpR (1st on call bleep 8131)

THE NATIONAL HOSPITAL FOR NEUROLOGY AND NEUROSURGERY JULY 2022

ENDOVASCULAR PROCEDURES

Supine position for 4-6 hours – *with leg straight at puncture site*

Neurological observations
GCS, Pupil, Limb assessment
Every 20 minutes for 2 hours
Every 30 minutes for further 2 hours
1 hourly for 2 hours
2 hourly for 6 hours
4 hourly thereafter if stable
(Duration dependent on procedure)

Puncture site
Check for swelling, bruising, bleeding - m*ild arterial / venous oozing possible*

Pedal pulses

Angiogram: Min 6h
Endovascular Procedure: Min 24h

FOLLOW POST-PROCEDURE INSTRUCTIONS in EPIC

FOLLOW ESCALATION PROTOCOL In event of **COMPLICATIONS** (***see poster)***

ESCALATION PROTOCOL
Post-Endovascular Intervention / Angiogram via GROIN Puncture or Post Insertion / Removal of a Vascular Access Device (e.g. Vascath)

University College London Hospitals NHS Foundation Trust

POST-PROCEDURE CARE

Supine position for 4-6 hours
with leg straight at puncture site

Patient can then mobilise **gently** but must remain under close observation for **specified period, dependent on procedure** – *if a patient insists on leaving the ward area during this specified period, they must be advised that it is against medical advice and this must be documented*
Consider need to extend duration of bed rest / observation if patient requires anticoagulation

Post-procedure checks

Neurological observations
GCS, Pupil, Limb assessment
Every 20 minutes for 2 hours
Every 30 minutes for further 2 hours
1 hourly for 2 hours
2 hourly for 6 hours
4 hourly thereafter if stable
(Duration dependent on procedure)

Puncture site
Check for swelling, bruising, bleeding - *mild arterial / venous oozing possible*

Pedal pulses

Angiogram: Min 6h
Endovascular Procedure: Min 24h

POSSIBLE COMPLICATIONS & EMERGENCY MANAGEMENT

HAEMATOMA / PSEUDOANEURYSM
- Swelling or pulsatile femoral mass and localised pain (if unsure if present compare with other side)
- Deficits in movement and sensation
- ***Apply manual pressure (minimum 10min) then FemoStop device***
- ***US +/- CTA/CTV*** **-** discuss with Neuroradiology SpR

RETROPERITONEAL BLEED
- Severe back / abdominal / groin pain *(Retroperitoneal haematoma may present with abdominal pain only)*
- Hypotension
- Bruising at puncture site, flanks or periumbilical area
- ***Apply manual pressure (minimum 10min) then FemoStop device***
- ***Venous gas*** *immediately to assess Hb drop / acidosis*
- ***Attending SpR*** *to decide if urgent CTA/V required - dual phase CT diaphragm to mid thigh + advise re: FemoStop removal for CT scan and re-application (Bleep 8251)*
- ***Team to remove FemoStop on CT scanner if stable cardiovascularly*** *- and re-apply immediately post scan*

ACTIVE ARTERIAL / VENOUS BLEEDING AT PUNCTURE SITE
To be differentiated from small oozing
- ***Apply manual pressure (minimum 10min) then FemoStop device - discuss with Neuroradiology SpR***

LIMB ISCHAEMIA
- Diminished pulses / coolness / pallor + changed sensation
- ***Discuss with Vascular Surgery SpR***

STROKE
- Neurological deficit, reduced consciousness
- ***Inform INR and team in charge of patient***
- ***Discuss urgent Imaging with Neuroradiology – CT/MRI***

ESCALATE – any **ONE** sign / symptom should activate alert

Escalate alert activated if routine post-procedure observations identify possible complications

E	Endovascular treatment / angiogram with groin puncture or insertion / removal of a vascular access device *Are any of the following signs / symptoms present?*
S	Swelling / haematoma / bleeding at puncture site
C	Cardiovascular changes – *hypotension, tachycardia, cool peripheries*
A	Abdominal **PAIN** / groin **PAIN** • *Retroperitoneal haematoma may present with only abdominal pain –* ***PAIN IS MOST COMMON SYMPTOM*** • *May also have low urine output / urinary retention* • *Sensation of 'POP' in abdomen*
L	Leg / thigh pain or swelling on side of groin puncture
A	Agitation or sweating
T	Tenderness at puncture site
E	Escalate if any **ONE** of above signs or symptoms present *Even if no outward sign of bleeding a* ***retroperitoneal haemorrhage*** *remains a possibility and should prompt the appropriate imaging*

APPLY MANUAL PRESSURE if required & IMMEDIATELY REFER TO:

- **Neurosurgery SpR** (Bleep 8100)
- **+/- Neurology SHO** (Bleep 8120) *if neurology patient*
- **+/- Stroke SpR** (Bleep 1145) *if stroke patient*
- **+ Outreach** (Bleep 8277)
- **+ Anaesthetic SpR** (Bleep 8131)

Neurosurgery SpR +/- Neurology / Stroke SpR should:
- Immediately attend and examine patient
- Continue manual pressure (minimum 10min) then FemoStop device
- Attending SpR to decide if urgent CTA/V required - dual phase CT diaphragm to mid thigh + advise re: FemoStop removal for CT scan and re-application post scan (Bleep 8251)

If abdominal bleed suspected –
- Immediate insertion of two wide bore cannulae
- Blood / fluid resuscitation +/- major haemorrhage protocol if haemodynamically compromised
- Send urgent full blood count, coagulation screen and x-match 4 units
- Arterial or venous gas to assess Hb and acidosis

If CT shows abdominal bleeding –
- Activate ***Pathway for Acute Vascular Bleeding***
- Message SAH / Vascular WhatsApp team
- Inform INR team
- Contact Duty Consultant Intensivist for Neurocritical Care

IF CT SHOWS ABDOMINAL BLEEDING ACTIVATE *'PATHWAY FOR ACUTE VASCULAR BLEEDING'*

ONGOING BLEEDING REQUIRING VASCULAR SURGICAL INPUT

Neurosurgery SpR to make immediate contact with Royal Free Hospital (RFL) Vascular Consultant via RFL switchboard:
0207 794 0500

1. **Contact on call Vascular Consultant** *If no response*
2. **Call 2nd on call Vascular Surgeon** *If no response*
3. **Call Clinical Director or Deputy Ext 32405**

Neurosurgery SpR should contact these people in order until they get through to someone in person. Leaving a telephone message is **not** sufficient as contact with the vascular team.

Consultant INR should take the lead in making subsequent contact with the consultant vascular surgeon to discuss case details. This can be delegated to the radiology SpR if consultant not immediately available.

Patient should be transferred to NHNN 1st Floor Neurocritical Care with low threshold to sedate and intubate to facilitate safe management.

Consultant intensivist will provide additional information to RFL vascular team regarding patient's overall condition and suitability for transfer to RFL.

IF PATIENT ACCEPTED FOR TRANSFER:
NHNN team arrange Blue Light Ambulance for urgent transfer to RFL Emergency Department

No need to wait for confirmation of ITU bed at RFL.

IF PATIENT TOO UNSTABLE FOR TRANSFER:
Vascular Surgeon to travel to NHNN

NHNN Theatre team mobilised to prepare for vascular case

(This information links into the RFL protocol shown in full in their PATHWAY FOR ACUTE VASCULAR BLEEDING IN REFERRING HOSPITAL)

THE NATIONAL HOSPITAL FOR NEUROLOGY AND NEUROSURGERY AUGUST 2023

P Grover / P Cowley / S Fairley – For review August 2024

CHECK OPERATION NOTE / POST-PROCEDURE INSTRUCTIONS

Check ***specific instructions*** *from Neurosurgeon / Neuroradiologist / Anaesthetist*

If no specific instructions use ***standard protocols***

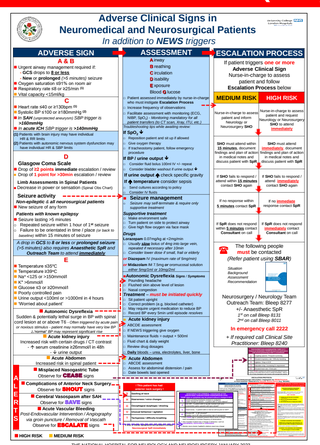
A B C D E assessment

WOUND ASSESSMENT

PAIN ASSESSMENT

If patient's condition causing concern:
1. Follow **NEWS** and **Adverse Clinical Signs** escalation protocols
1. Increase frequency of observations
2. Inform nurse in charge and medical staff

Follow escalation protocol for ADVERSE CLINICAL SIGNS TRIGGERS (*see poster*)



THE NATIONAL HOSPITAL FOR NEUROLOGY AND NEUROSURGERY JANUARY 2023

POST-OP FREQUENCY OF OBSERVATIONS
Highest risk of bleeding in first 6 hours

→ **Every 20 minutes for 2h** → **½ hourly for 2h** → **1 hourly for 2h** → **2 hourly for minimum 6h** → **If overnight continue 2 hourly until 0800** → **4 hourly thereafter if stable**